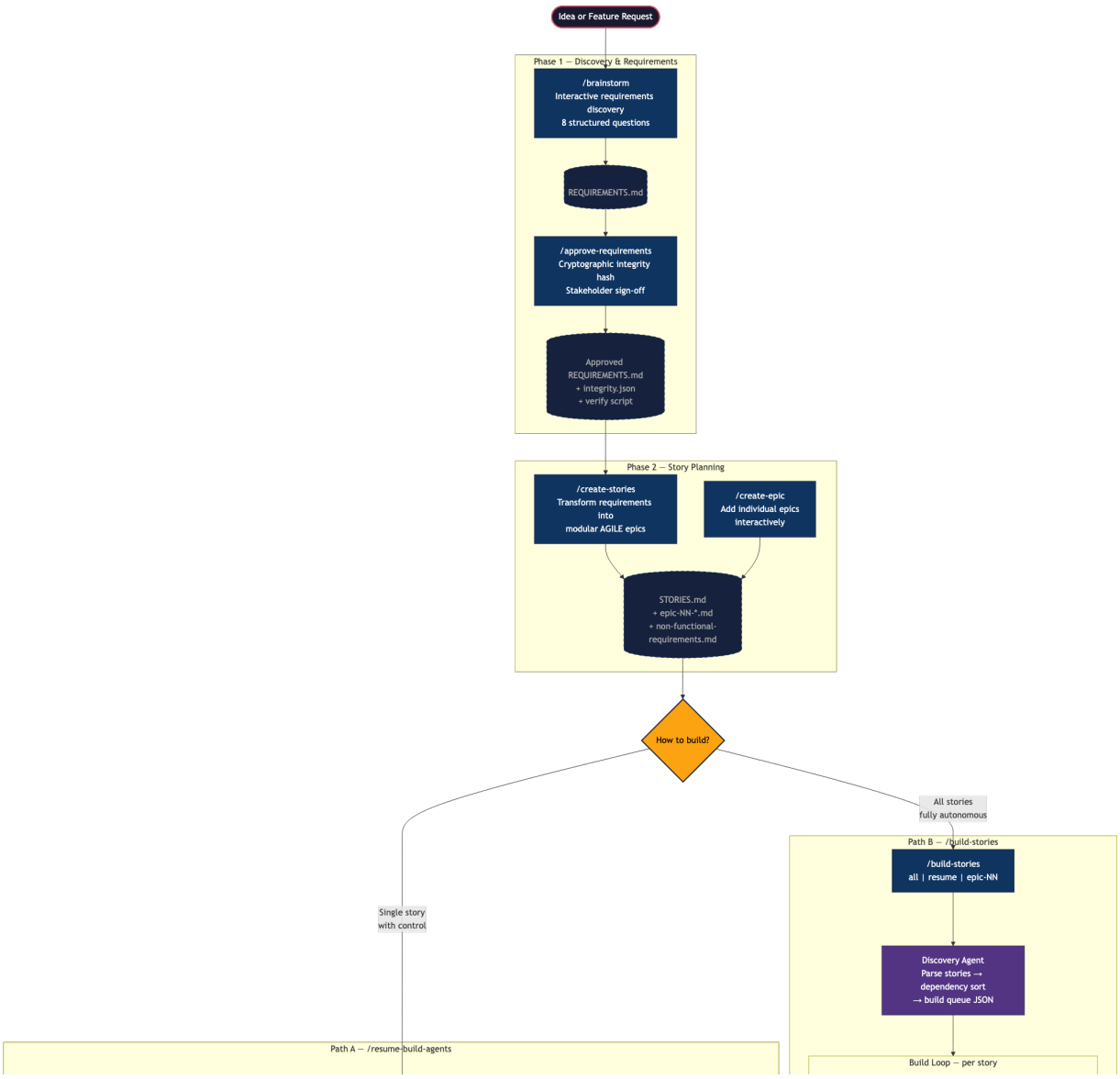
Automated Design, Develop, Test Workflow

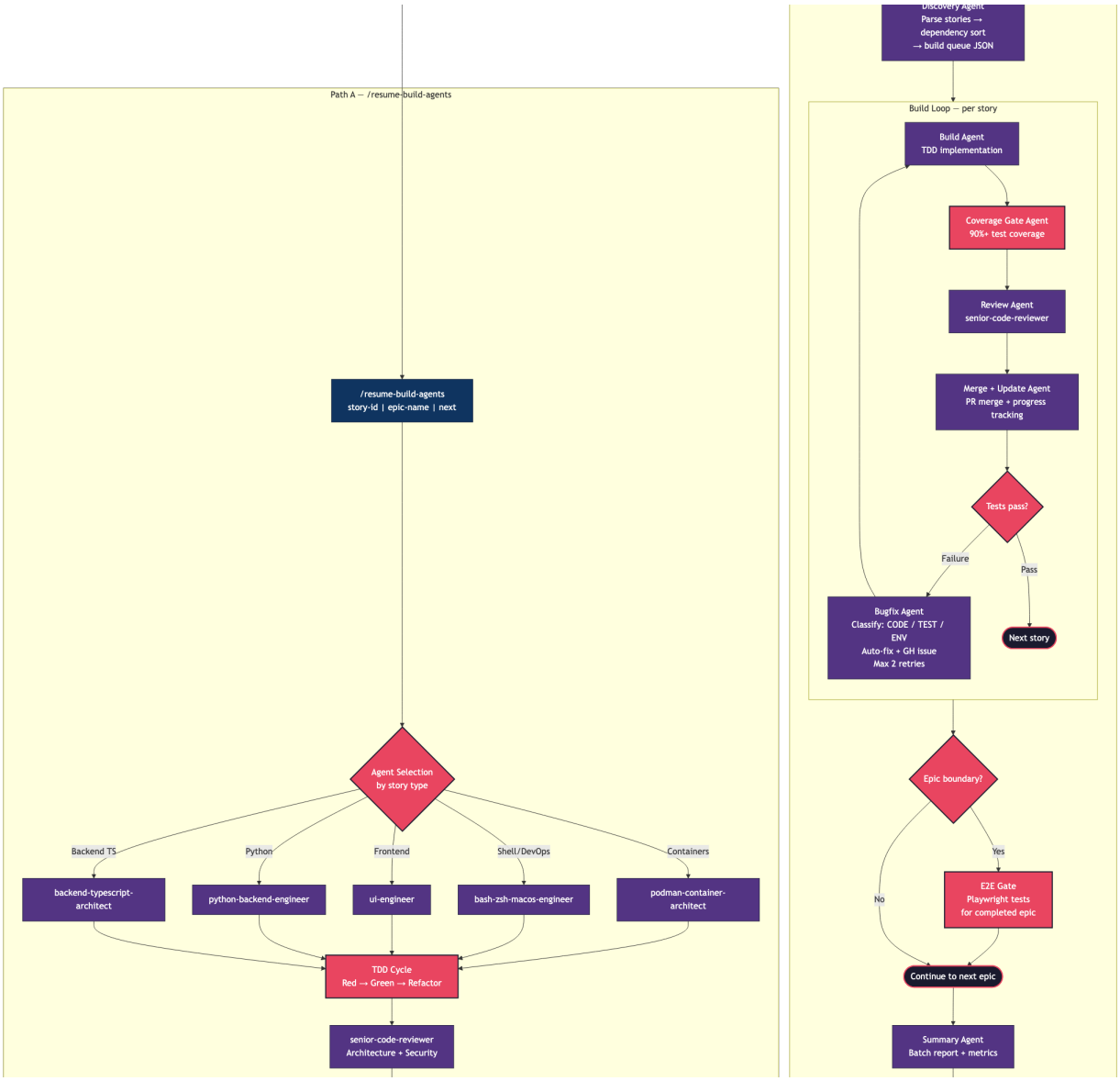
End-to-end workflow from idea to deployed, tested, reviewed code using Claude Code's multi-agent orchestration system

Workflow Diagram

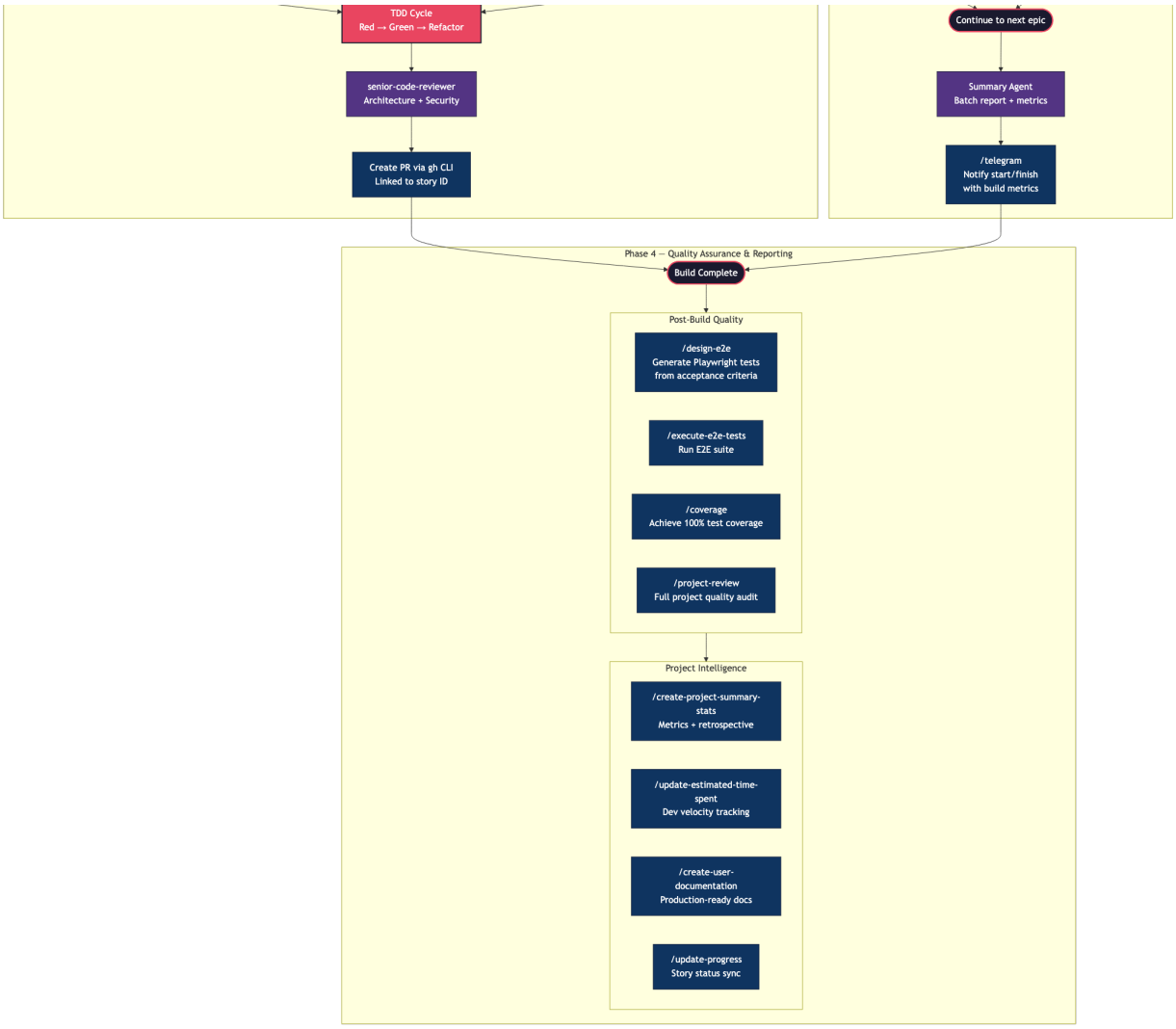
Phases 1-2: Discovery, Requirements & Story Planning



Phase 3: Build Paths (Controlled & Autonomous)



Phase 3-4: Build Loop, E2E Gate & Quality Reporting



Phase 1 — Discovery & Requirements

Step	Command	What happens
1	`/brainstorm`	Interactive discovery session: 8 structured questions covering problem space, personas, success metrics, capabilities, scope boundaries, technical constraints, priority, and acceptance criteria. Produces REQUIREMENTS.md.

Alternative: Use `/create-epic` to add individual epics interactively without going through full requirements discovery.

Phase 2 — Story Planning

Step	Command	What happens
2	`/create-stories`	Transforms REQUIREMENTS.md into modular AGILE structure: STORIES.md overview, individual epic-NN-*.md files with INVEST-compliant user stories, and non-functional-requirements.md.
2b	`/create-epic` *(optional)*	Adds individual epics interactively with 8-question discovery flow. Generates properly numbered stories following {Epic}.{Feature}-{NNN} format.

Phase 3 — Build

Two paths depending on desired control level:

Path A: `/resume-build-agents` (Controlled)

For building **one story at a time** with visibility into agent selection and review.

```
/resume-build-agents <story-id | epic-name | next> [--skip-review] [--no-tests]
```

1. Validates environment (clean git, STORIES.md exists, GitHub auth)
2. Auto-selects specialized agent based on story type and tech stack
3. Creates feature branch (feature/$STORY_ID)
4. Runs TDD cycle (Red, Green, Refactor)
5. Mandatory code review via senior-code-reviewer
6. Creates PR linked to story ID

Available agents: backend-typescript-architect, python-backend-engineer, ui-engineer, bash-zsh-macos-engineer, podman-container-architect, qa-engineer

Path B: `/build-stories` (Fully Autonomous)

For building **all incomplete stories** across epics with automated error recovery.

```
/build-stories [all | resume | epic-NN | epic-name] [--dry-run] [--auto]
[--skip-coverage] [--e2e-gate=block|warn|off]
```

1. **Discovery Agent** — parses stories, resolves dependencies via topological sort, produces build queue
2. **Build Agent** — TDD implementation using the appropriate specialized agent
3. **Coverage Gate** — enforces 90%+ test coverage, adds missing tests
4. **Review Agent** — senior-code-reviewer validates architecture, security, performance
5. **Merge + Update Agent** — merges PR, updates progress tracking
6. **Bugfix Loop** — on failure, classifies as CODE_BUG / TEST_BUG / ENV_ISSUE, creates GitHub issue, auto-fixes (max 2 retries)
7. **E2E Gate** — runs Playwright tests at epic boundaries
8. **Summary Agent** — generates batch report with metrics
9. **Telegram notification** — posts start/finish with build metrics

Progress tracking: docs/stories/.build-progress.md maintains per-story status (DONE / IN_PROGRESS / FAILED / SKIPPED / PENDING).

Phase 4 — Quality Assurance & Reporting

Post-build commands for additional quality gates and project intelligence:

Command	Purpose
`/design-e2e`	Generate Playwright E2E tests from acceptance criteria
`/execute-e2e-tests`	Run the E2E test suite
`/coverage`	Analyze and fill test coverage gaps
`/project-review`	Full project quality audit with scoring
`/create-project-summary-stats`	Generate metrics and retrospective
`/update-estimated-time-spent`	Track development velocity
`/create-user-documentation`	Generate production-ready docs
`/update-progress`	Sync story status across files

Supporting Commands

Category	Command	Purpose
Issues	`/create-issue`	Create comprehensive GitHub issues with defect analysis
Issues	`/fix-github-issue`	Investigate and fix issues from GitHub
Quality	`/roast`	Brutal honest code assessment
DevOps	`/check-releases`	Monitor upstream dependency updates
DevOps	`/plan-release-update`	Plan Nix-based release updates
Generators	`/create-command`	Scaffold new slash commands
Generators	`/create-skill`	Scaffold new skills
Generators	`/create-agent`	Scaffold new agent definitions

Agent Roster

Agent	Specialization
backend-typescript-architect	Bun runtime, advanced TypeScript, microservices
python-backend-engineer	FastAPI, uv, SQLAlchemy, async Python
ui-engineer	Modern frontend, component architecture, responsive design
bash-zsh-macos-engineer	macOS shell scripting, automation, CI/CD
podman-container-architect	OCI containers, multi-stage builds, rootless Podman
qa-engineer	Test strategy, quality metrics, defect management
senior-code-reviewer	Architecture validation, security audits, best practices
meta-agent	Generates new agent definitions

Quick Reference

```
# Full workflow: idea -> deployed code
/brainstorm # 1. Discover requirements
/create-stories # 2. Generate stories

# Then choose your build path:
/resume-build-agents next # A. One story at a time (controlled)
/build-stories all # B. All stories (autonomous)

# Post-build quality:
/design-e2e epic-01 # Generate E2E tests
/coverage # Fill coverage gaps
/create-project-summary-stats # Generate retrospective
```

Annex — Repository File Structure

Complete layout of the claude-code-config repository. All paths are relative to the repository root.

claude-code-config/	
CLAUDE.md	Global instructions for Claude Code
README.md	Setup and overview
WORKFLOW.md	This workflow document
install.sh	Portable installer (symlinks + MCP config)
settings.json	Editor settings (statusline, plugins)
keybindings.json	Keyboard shortcuts
statusline-command.sh	Statusline display script
agents/	12 agent definitions (flat .md files)
backend-typescript-architect.md	Bun + TypeScript backend
bash-zsh-macos-engineer.md	macOS shell automation
crypto-coin-analyzer.md	Single crypto ticker analysis
crypto-market-agent.md	Crypto market data retrieval
executive-summary-generator.md	Client company summaries
meta-agent.md	Generates new agent definitions
podman-container-architect.md	OCI containers, Podman
professional-profile-researcher.md	LinkedIn/profile research
python-backend-engineer.md	FastAPI + uv + Python
qa-engineer.md	Testing strategy and QA
senior-code-reviewer.md	Architecture/security review
ui-engineer.md	Frontend components and UI
commands/	17 slash commands by category
dev/	
brainstorm.md	Interactive requirements discovery
create-todo.md	Transform plans into atomic tasks
devops/	
check-releases.md	Monitor upstream dependency updates
plan-release-update.md	Plan Nix-based release updates
issues/	
create-issue.md	Create GitHub issues with defect analysis
fix-github-issue.md	Investigate and fix GitHub issues
project/	
create-project-summary-stats.md	Metrics and retrospectives
create-user-documentation.md	Production-ready docs
sync-progress.md	Progress synchronization
update-estimated-time-spent.md	Dev velocity tracking
update-progress.md	Manual progress updates
quality/	
coverage.md	Test coverage analysis
project-review.md	Full project quality audit
roast.md	Brutal honest code assessment
research/	
client-analysis.md	Executive summaries
crypto-analysis.md	Crypto market analysis
profile-analysis.md	Professional profile research
skills/	13 skills (SKILL.md + supporting prompts)
build-stories/	Batch build orchestrator
SKILL.md	Main dispatcher
discovery-agent-prompt.md	Story parsing + dependency sort
bugfix-agent-prompt.md	Root cause diagnosis + auto-fix
coverage-gate-prompt.md	Test coverage analysis

e2e-gate.md	Epic-boundary E2E testing
merge-update-prompt.md	PR merge + progress tracking
summary-prompt.md	Batch summary report
batch-progress.md	Progress file format
story-parser.md	Story extraction + validation
dependency-resolver.md	Topological sort + priority
resume-build-agents/	Single-story builder
SKILL.md	Main orchestrator
agent-coordination.md	Agent selection logic
quality-gates.md	TDD discipline + test requirements
create-stories/	Requirements to stories generator
SKILL.md	Main generator
generation-rules.md	Story generation rules
claude-md-update.md	CLAUDE.md update template
create-epic/	Interactive epic creation
SKILL.md	Epic generator with INVEST criteria
design-e2e/	Playwright E2E test generation
SKILL.md	Test design + execution
generation-rules.md	Test generation rules
bugfix-prompt.md	E2E test bugfix loop
execute-e2e-tests/	E2E test runner
SKILL.md	Test execution
instructions.md	Runner instructions
interactive-questions.md	Configuration questions
approve-requirements/	Cryptographic approval
SKILL.md	Approval process
approval-process.md	Initial approval flow
change-control.md	Re-approval flow
telegram/	Telegram Bot messenger
SKILL.md	Bot API integration
instructions.md	Message formatting rules
claude-docs/	Documentation viewer
SKILL.md	Doc viewer skill
instructions.md	Viewer instructions
create-agent/	Agent scaffolder
SKILL.md	Generator skill
generation-rules.md	Agent generation rules
interactive-questions.md	Discovery questions
create-command/	Command scaffolder
SKILL.md	Generator skill
generation-rules.md	Command generation rules
interactive-questions.md	Discovery questions
create-skill/	Skill scaffolder
SKILL.md	Generator skill
generation-rules.md	Skill generation rules
interactive-questions.md	Discovery questions
demo/	Demo skill
SKILL.md	Minimal example
demo-rules.md	Demo conventions
templates/	Shared reference templates
agent-template.md	Agent definition template
command-template.md	Command definition template
skill-template.md	Skill definition template
reference-docs/	Context references for Claude
python.md	Python conventions
source-control.md	Git/source control practices
docker-uv.md	Container + uv tooling

docs/	User-facing documentation
generators.md	Generator skills documentation
hooks/	Hook configurations
README.md	Hook setup guide
mcp/	MCP server configuration
config.template.json	MCP template (env var substitution)